

CHERNI VRAH, Bulgaria

WMO: 156130

Lat: 42.562N

Lon: 23.285E

Elev: 7520

StdP: 11.12

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.5	1.0	-13.7	3.5	10.6	-8.3	4.7	12.4	76.4	19.9	75.2	20.7	27.1	320	1.029

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.0	63.2	51.7	60.6	51.0	58.4	50.2	54.3	59.2	52.8	57.6	51.5	56.2	12.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.4	77.6	56.3	50.8	73.1	54.8	49.3	69.1	53.4	26.0	59.1	24.9	57.8	24.0	56.6	65.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
63.3	53.9	44.4	-7.0	68.9	4.3	2.3	-10.1	70.6	-12.6	71.9	-15.0	73.2	-18.2	74.9		
			-7.3	59.8	4.4	2.7	-10.5	61.7	-13.0	63.2	-15.5	64.7	-18.7	66.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.5	19.1	18.1	22.3	28.9	38.9	45.6	49.2	49.9	42.4	36.1	29.0	21.5	
	DBStd	13.63	8.53	8.80	8.37	7.05	6.81	6.93	6.19	5.65	6.75	7.89	9.41	8.47	
	HDD50	6207	957	892	860	634	348	166	90	72	240	432	632	884	
	HDD65	11495	1422	1312	1325	1084	810	582	489	469	677	895	1081	1349	
	CDD50	186	0	0	0	0	3	33	66	68	13	2	1	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	18.6	22.4	23.2	23.4	21.2	17.1	14.7	14.1	13.1	14.8	17.9	20.4	21.6	
Precipitation	PrecAvg	28.3	1.9	1.8	2.1	2.4	3.5	3.3	2.7	2.4	2.4	2.2	1.6	1.9	
	PrecMax	45.1	3.9	3.0	4.9	5.7	7.4	6.8	6.0	7.7	8.0	4.6	3.9	3.6	
	PrecMin	16.1	0.5	0.5	0.7	0.6	1.3	0.9	0.8	0.2	0.5	0.4	0.1	0.6	
	PrecStd	6.6	0.9	0.7	0.9	1.1	1.6	1.5	1.5	1.6	1.6	1.2	0.8	0.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	37.0	42.8	48.7	57.3	64.3	68.0	66.0	59.9	55.4	52.0	40.5	
		MCWB	29.3	28.5	35.7	42.1	47.5	52.8	51.3	52.6	49.8	46.0	39.0	31.5	
	2%	DB	34.6	33.2	37.5	43.1	53.5	60.0	63.4	63.0	56.7	51.1	46.6	37.4	
		MCWB	27.4	28.2	32.0	36.6	44.6	50.7	51.5	51.8	48.6	43.4	35.2	30.6	
	5%	DB	31.5	30.7	34.7	40.0	50.7	57.6	60.7	60.5	54.4	48.7	43.0	34.2	
		MCWB	27.4	27.7	30.6	35.6	43.9	50.0	51.2	51.0	47.0	41.5	33.8	30.3	
	10%	DB	29.3	28.6	32.7	37.3	48.0	55.1	58.2	58.1	51.6	46.1	39.8	31.6	
		MCWB	26.9	26.9	30.3	34.4	42.7	48.8	50.4	50.0	46.0	40.2	33.7	28.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.4	31.8	36.7	42.3	49.0	55.9	57.0	57.0	51.6	48.6	42.0	36.5
			MCDB	34.4	32.7	40.8	47.9	54.4	60.9	62.2	62.1	57.2	52.9	47.7	37.2
2%		WB	30.3	29.7	33.6	38.3	46.5	53.1	54.2	54.2	49.9	45.9	39.3	33.7	
		MCDB	31.7	31.1	35.1	41.1	51.0	57.6	59.0	59.3	55.0	49.5	43.0	34.7	
5%		WB	28.6	28.3	31.9	36.3	45.1	51.3	52.7	52.6	48.4	43.9	37.2	31.4	
		MCDB	30.1	29.9	33.4	38.8	48.9	55.8	57.4	57.8	52.6	46.7	39.1	32.8	
10%		WB	27.0	26.9	30.5	34.6	43.7	49.5	51.4	51.1	46.7	42.2	35.1	29.7	
		MCDB	28.6	28.3	32.4	36.9	47.0	53.9	56.0	56.1	50.4	44.3	37.9	31.1	
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.5	7.7	6.9	7.2	8.0	8.3	8.0	7.4	7.4	7.1	7.1	
		MCDBR	8.3	7.9	7.7	9.2	10.1	10.5	11.4	10.9	10.0	9.1	8.9	7.8	
		MCWBR	6.6	6.4	5.7	6.6	7.1	8.5	8.2	7.6	7.1	7.2	6.1	6.2	
	5% WB	MCDBR	7.0	7.8	6.9	8.2	9.1	9.0	9.8	9.8	9.4	7.8	8.2	6.2	
		MCWBR	6.5	7.3	6.1	6.4	7.1	8.7	8.3	7.8	7.2	6.7	6.5	5.9	
Clear-Sky Solar Irradiance	taub	0.241	0.260	0.290	0.329	0.325	0.322	0.323	0.329	0.312	0.285	0.262	0.243		
	taud	2.492	2.478	2.461	2.365	2.408	2.429	2.417	2.398	2.449	2.531	2.569	2.528		
	Ebn at Noon	295	305	305	298	300	300	299	294	293	291	285	285		
	Edh at Noon	22	27	31	37	36	36	36	35	31	25	21	20		
All-Sky Solar Radiation	RadAvg	515	776	1096	1407	1678	1906	1978	1789	1319	920	591	434		
	RadStd	77	100	120	169	139	169	155	159	108	114	60	63		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	-261	N/A	N/A

Nomenclature: See separate page